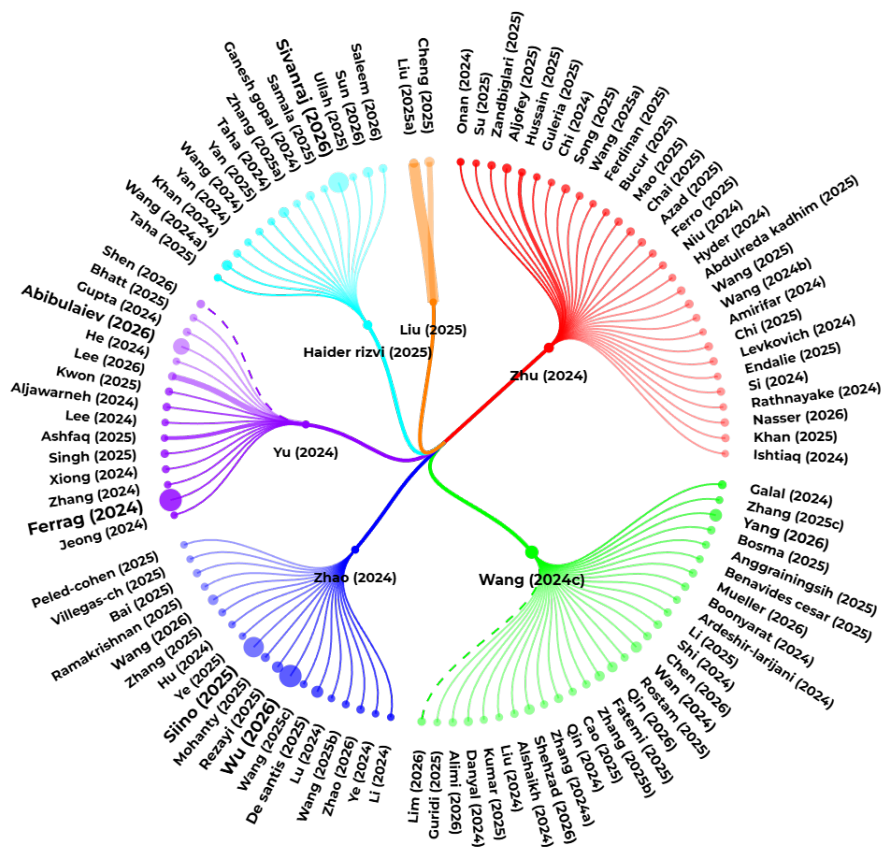


Literature Review and research themes

Your query

[TITLE=("embedding techniques" OR "tf idf" OR ("bert" OR "Word Embeddings" OR "Text Classification" OR "Contextualized Embeddings" OR "Knowledge Graph Embeddings" OR "NLP")) AND DOMAIN=(COMP) AND RANKING=(4*) AND YEAR=(2024-2026)]

Star Map



Number of documents: 119 (threshold: 1.2879256965944272) | Year range: 2020 - 2026

Theme 1 / Diverse NLP Applications

Summary:

This theme encompasses a wide range of topics within the field of Natural Language Processing (NLP) and its applications. The titles suggest a strong focus on text classification, sentiment analysis, anomaly detection, and the use of BERT and its variants for enhancing the performance of NLP tasks. There are also titles related to the application of NLP in specific domains such as medical text classification, construction contract analysis, and cultural heritage.

Synthesis of all articles in this group:

The cluster of Diverse NLP Applications encompasses various subthemes, including Text Classification, Anomaly Detection and Cybersecurity, and NLP Applications in Specific Domains. According to Zhu (2024), recent studies in text classification have explored approaches such as prompt-learning and optimized DistilBERT to improve performance. Khan (2025) proposed a novel approach for anomaly detection in attributed networks, while Nasser (2026) applied NLP in the construction domain to analyze contract selection factors. The use of BERT-based models has been effective in addressing various challenges, including text classification (Hussain, 2025), anomaly detection (Khan, 2025), and extractive summarization (Wang, 2024). These studies demonstrate the diversity of NLP applications and the significance of developing robust and accurate models, as highlighted by Chi (2025) and Ferro (2025). Overall, the cluster showcases the potential of NLP in various domains, including healthcare, social media, construction, and cultural heritage, as seen in the works of Zandbiglari (2025), Levkovich (2024), and Onan (2024).

Subthemes:

• **Text Classification :** (Zhu Y. (2024) [9], Aljofey A. (2025) [10], Song R. (2025) [11], Wang Y. (2025) [12], Chi D. (2025) [13], Abdulreda Kadhim E. (2025) [14], Chai X. (2025) [15], Mao X. (2025) [19], Azad M. (2025) [20], Hussain A. (2025) [25], Wang B. (2025) [26], Bucur A. (2025) [30], Rathnayake H. (2024) [31], Guleria P. (2025) [32], Su S. (2025) [33], Endalie D. (2025) [34], Chi Z. (2024) [36], Ishtiaq A. (2024) [37], Amirifar T. (2024) [38], Ferdinan T. (2025) [35])

Recent studies in text classification have explored various approaches to improve performance, including prompt-learning (Zhu, 2024), optimized DistilBERT (Aljofey, 2025), and causal keyword driven reliable text classification (Song, 2025). Soft prompt learning methods have also been investigated, such as simulating human cognitive capabilities for medical text classification (Wang, 2025). Additionally, researchers have proposed models combining BERT with other techniques, like TCN and BiLSTM, for sentiment analysis (Chi, 2025) and text summarization (Abdulreda, 2025). These studies demonstrate the effectiveness of these approaches in achieving high accuracy and F1 scores. Furthermore, log sequence anomaly detection based on template and parameter parsing via BERT has also been explored (Chai, 2025). Overall, these studies highlight the importance of developing robust and accurate text classification models, and the potential of using advanced techniques like BERT and prompt-learning to improve performance. The provided citations focus on various aspects of text classification, including the use of BERT and RoBERTa models for ransomware detection (Hussain, 2025), gaze-aware graph convolutional networks for text classification (Wang, 2025), and NLP approaches to modeling depression in social media (Bucur, 2025). Mao (2025) proposes an enhanced representation learning and

generalization model for English text classification, while Azad (2025) explores the self-assessment of depression for Bangladeshi university students using machine learning and NLP. Rathnayake (2024) presents an AdapterFusion-based multi-task learning approach for code-mixed and code-switched text classification, and Guleria (2025) proposes an NLP-based text classification using TF-IDF enabled fine-tuned long short-term memory. These studies demonstrate the diversity of text classification applications and the effectiveness of various machine learning and NLP techniques in addressing these tasks. The use of transfer learning-based models, such as BERT and RoBERTa, has shown promising results in detecting ransomware and classifying text (Hussain, 2025). Additionally, the integration of gaze-aware graph convolutional networks has improved text classification performance (Wang, 2025). The analysis of depression in social media using NLP approaches has also been explored, highlighting the importance of considering ethical issues in collecting and processing mental health data (Bucur, 2025). Overall, these studies contribute to the advancement of text classification techniques and their applications in various domains. The citations in this chunk focus on various aspects of text classification, including the use of attention feature enhancement networks (Su, 2025), fine-tuning BERT models for multiclass Amharic news document categorization (Endalie, 2025), and exploring whether pretrained English language models can benefit non-English NLP systems in low-resource scenarios (Chi, 2024). Additionally, the citations discuss the use of novel transformer-based BERT embedding and class probability features for product helpfulness detection (Ishtiaq, 2024), an NLP-deep learning approach for product rating prediction based on online reviews and product features (Amirifar, 2024), and fortifying NLP models against poisoning attacks using personalized prediction architectures (Ferdinan, 2025). These studies demonstrate the effectiveness of various techniques in improving text classification accuracy and robustness, such as fine-tuning pretrained models (Endalie, 2025), using cross-lingual knowledge transfer methods (Chi, 2024), and employing personalized model architectures (Ferdinan, 2025). The results show that these approaches can achieve high accuracy, precision, and recall, and outperform baseline models in various tasks, including text classification (Su, 2025), sentiment analysis (Ishtiaq, 2024), and product rating prediction (Amirifar, 2024).

• **Anomaly Detection and Cybersecurity :** (Khan W. (2025) [17], Zandbiglari K. (2025) [18], Levkovich I. (2024) [21], Si J. (2024) [22], Niu H. (2024) [23], Wang Y. (2024) [24], Hyder S. (2024) [28], Onan A. (2024) [29])

The subtheme of Anomaly Detection and Cybersecurity involves various techniques to identify unusual patterns in data, as seen in Khan (2025), who proposed a novel approach for anomaly detection in attributed networks using proximity preservation and advanced embedding techniques. This method effectively preserves both node features and network structure, making the embedding robust and distinguishable. In the context of healthcare, Zandbiglari (2025) developed a multi-label NLP framework with transformer models and semantic retrieval-based annotation to detect suicidal behaviors in Electronic Health Records (EHRs). The proposed approach captures the complexity of suicidal behaviors more effectively than traditional binary classification. Levkovich (2024) evaluated the use of BERT-based and large language models for suicide detection, prevention, and risk assessment, demonstrating their potential for identifying and detecting suicidal behaviors. Other studies, such as Si (2024) and Niu (2024), utilized BERT-based models for trajectory recovery and anomaly detection in EHRs, respectively. The use of BERT-based models, as seen in Wang (2024) and Hyder (2024), also extends to extractive summarization and deceptive review detection in social media. These studies highlight the significance of anomaly detection and cybersecurity in various domains, including healthcare and social media, and the effectiveness of BERT-based models in addressing these challenges. FuzzyTP-BERT enhances text summarization (Onan, 2024)

• **NLP Applications in Specific Domains :** (Nasser M. (2026) [16], Ferro S. (2025) [27])
NLP is applied in construction and cultural heritage domains, as seen in Nasser (2026) and Ferro (2025), to analyze contract selection factors and enrich artefact descriptions,

respectively, using semantic analysis and NLP-driven approaches.

Theme 2 / BERT Applications and Enhancements

Summary:

This theme encompasses a wide range of topics related to natural language processing (NLP), with a significant focus on the application and enhancement of BERT models. The cluster includes surveys, applications, and challenges related to BERT in various domains such as information retrieval, sentiment analysis, named entity recognition, text classification, and more.

Synthesis of all articles in this group:

The cluster 'BERT Applications and Enhancements' encompasses a wide range of applications and enhancements of BERT in natural language processing (NLP) tasks. According to Wang (2024), BERT-based methods have been extensively explored for information retrieval, highlighting the advantages of employing encoder-based BERT models. Yang (2026) and Wan (2024) propose methods for named entity recognition and sentiment analysis, respectively, demonstrating the effectiveness of BERT in various NLP tasks. Shehzad (2026) presents a multimodal learning framework for visual question answering, combining BERT with vision transformers. Qin (2024) proposes an adversarial fine-tuning strategy to boost the generalization of fine-tuning BERT for fake news detection. Galal (2024) rethinks the way of using BERT sentence embedding for text classification, proposing different aggregation architectures and approaches. Chen (2026) introduces ASDE, a low-budget text classification method via active semi-supervised learning with debiasing training mechanism. Alimi (2026), Bosma (2025), Zhang (2025), Alshaikh (2024), Boonyarat (2024), Cao (2025), and Li (2025) also demonstrate the effectiveness of BERT in various NLP tasks, including named entity recognition, text classification, sentiment analysis, and intrusion detection. Furthermore, Zhang (2024), Kumar (2025), and Anggrainingsih (2025) highlight the effectiveness of BERT-based models in handling complex contexts and ambiguities. Overall, the cluster highlights the versatility and effectiveness of BERT in various NLP tasks, with applications in text classification, sentiment analysis, named entity recognition, and more.

Subthemes:

• **BERT Applications in NLP :** (Wang J. (2024) [39], Yang X. (2026) [40], Wan B. (2024) [41], Shehzad F. (2026) [43], Qin S. (2024) [46], Galal O. (2024) [47], Chen Y. (2026) [49], Alimi T. (2026) [51], Bosma J. (2025) [52], Zhang Y. (2025) [53], Alshaikh K. (2024) [54], Boonyarat P. (2024) [55], Cao Z. (2025) [56], Li S. (2025) [57], Zhang L. (2024) [59], Kumar A. (2025) [60], Anggrainingsih R. (2025) [61])

The applications of BERT in NLP have been extensively explored in recent years. Wang et al. (2024) provide a comprehensive survey of BERT-based methods for information retrieval, highlighting the advantages of employing encoder-based BERT models. Yang et al. (2026) propose a method for named entity recognition in traditional Chinese medical cases by integrating BERT and BiLSTM. Wan et al. (2024) introduce an Emotion-Cognitive Reasoning integrated BERT (ECR-BERT) for sentiment analysis of online public opinions on emergencies, which outperforms other BERT-based models. Shehzad et al. (2026) present an efficient multimodal learning framework for visual question answering on peripheral blood cells, combining BERT with vision transformers. Qin et al. (2024) propose an adversarial fine-tuning strategy to boost the generalization

of fine-tuning BERT for fake news detection. Galal et al. (2024) rethink the way of using BERT sentence embedding for text classification, proposing different aggregation architectures and approaches for using BERT as a feature extractor. Chen et al. (2026) introduce ASDE, a low-budget text classification method via active semi-supervised learning with debiasing training mechanism. These studies demonstrate the versatility and effectiveness of BERT in various NLP tasks. The applications of BERT in NLP have been explored in various studies. Alimi (2026) proposed ARDIAL-BERT for advancing multidialectal Arabic named entity recognition through continual pretraining. Bosma (2025) introduced the DRAGON benchmark for clinical NLP, which facilitates automated, large-scale, cost-effective data annotation. Zhang (2025) presented an enhanced few-shot text classification approach by integrating topic modeling and prompt-tuning. Alshaikh (2024) developed a BERT-based model for aspect-based sentiment analysis for analyzing Arabic open-ended survey responses. Boonyarat (2024) leveraged enhanced BERT models for detecting suicidal ideation in Thai social media content amidst COVID-19. Cao (2025) proposed KD-BERT, a lightweight knowledge distillation bidirectional encoder representations from transformers for IoT network intrusion detection. Li (2025) used knowledge distillation from BERT to CNN-BiLSTM for efficient intrusion detection for in-vehicle networks. These studies demonstrate the effectiveness of BERT in various NLP tasks, including named entity recognition, text classification, sentiment analysis, and intrusion detection. BERT-based models are effective in various NLP tasks, such as named entity recognition (Zhang, 2024), sentiment analysis (Kumar, 2025), and detecting misinformation (Anggrainingsih, 2025). These models can handle complex contexts and ambiguities, and their performance can be enhanced with techniques like multi-granularity adapters and efficient global pointers (Zhang, 2024). The quality of text embeddings generated by BERT-based models is also significant, with minimal differences between models like RoBERTa, BERT, and DistilBERT (Anggrainingsih, 2025).

- **Text Classification :** (Fatemi S. (2025) [44], Danyal M. (2024) [45], Shi S. (2024) [48], Mueller A. (2026) [50], Rostam Z. (2025) [58], Zhang Y. (2025) [63])
Summary unavailable for this subtheme.

- **Explainability and Interpretability :** (Ardeshtir-Larijani E. (2024) [62], Lim H. (2026) [4])
Summary unavailable for this subtheme.

- **Domain-Specific Applications :** (Benavides Cesar L. (2025) [65])
Summary unavailable for this subtheme.

Theme 3 / NLP in Various Domains

Summary:

This theme encompasses a wide range of topics primarily focused on natural language processing (NLP) and its applications across various domains, including healthcare, law, agriculture, education, and wireless communication. The titles suggest a strong emphasis on the development and application of large language models, BERT, and other deep learning frameworks for tasks such as text classification, entity alignment, sentiment analysis, and data generation.

Synthesis of all articles in this group:

The cluster 'NLP in Various Domains' encompasses a wide range of applications of natural language processing (NLP) across different fields, including healthcare, law, agriculture, education, and data generation (Wu, 2026; Siino, 2025; Villegas-Ch, 2025;

Mohanty, 2025). In healthcare, NLP has been used for neurological signal processing, text classification, and medical text classification, demonstrating its potential in improving healthcare outcomes (Wu, 2026; De Santis, 2025; Wang Y., 2026). The use of BERT and Transformer models has also been explored in various fields, including wireless communication, cross-lingual temporal knowledge graphs, and social media data (Zhao, 2026; Bai, 2025; Ramakrishnan, 2025). Additionally, advancements have been made in text classification and entity alignment, with applications in various domains (Wang, 2025; Lu, 2024; Li, 2024; Ye, 2024). NLP has also been integrated with other fields, such as education, where it has been used for automated semantic assessment and adaptive tutoring (Villegas-Ch, 2025). Furthermore, NLP has been used for data generation, including synthetic data generation using TimeGAN (Mohanty, 2025). Overall, the cluster highlights the versatility and effectiveness of NLP in different domains, and demonstrates the need for further research in this area (Wu, 2026; Siino, 2025; Villegas-Ch, 2025; Mohanty, 2025).

Subthemes:

• **Healthcare Applications :** (Wu D. (2026) [67], De Santis E. (2025) [69], Wang Y. (2026) [71], Wang H. (2025) [76], Peled-Cohen L. (2025) [82])

The citations provided focus on various healthcare applications, including neurological signal processing (Wu, 2026), text classification in healthcare discussions on social media (De Santis, 2025), medical text classification (Wang Y., 2026), and labeling tools for medical text classification (Wang H., 2025). These studies demonstrate the potential of deep learning and natural language processing (NLP) in improving healthcare outcomes. For instance, Neuro-BERT (Wu, 2026) uses self-supervised pre-training to improve downstream neurological-related tasks. Similarly, a comparative study (De Santis, 2025) shows that transformers outperform traditional methods for text classification in healthcare discussions on social media. Additionally, a systematic review (Peled-Cohen, 2025) highlights the opportunities and challenges of applying NLP to dementia research, including dementia detection, linguistic biomarker extraction, and caregiver support. The use of generative class feature fusion with agent attention (Wang Y., 2026) and interactive knowledge-assisted labeling tools (Wang H., 2025) also demonstrates the potential of NLP in medical text classification. Overall, these studies demonstrate the importance of NLP and deep learning in healthcare applications, and highlight the need for further research in this area.

• **Legal and Agricultural NLP :** (Siino M. (2025) [68], Rezayi S. (2025) [70])
Siino (2025) and Rezayi (2025) explore NLP applications in law and agriculture. Siino (2025) identifies trends in legal NLP, while Rezayi (2025) investigates large language models for food applications.

• **BERT and Transformer Models :** (Zhao Z. (2026) [73], Bai L. (2025) [74], Ramakrishnan S. (2025) [77], Hu Y. (2024) [80])
Recent studies have explored the applications of BERT and Transformer models in various fields. Zhao (2026) proposed a BERT-inspired framework for efficient CSI prediction and classification in wireless communication and sensing. Bai (2025) used BERT for joint multi-feature information entity alignment in cross-lingual temporal knowledge graphs. Ramakrishnan (2025) improved multi-label emotion classification on imbalanced social media data using BERT and clipped asymmetric loss, achieving notable improvements in macro-average F1-scores. Hu (2024) developed a BERT-based framework for recovering missing data segments in time-series load profiles. These studies demonstrate the versatility and effectiveness of BERT and Transformer models in different domains.

• **Text Classification and Entity Alignment :** (Wang M. (2025) [79], Lu Y. (2024) [81], Li F. (2024) [85], Ye H. (2024) [86])
Wang (2025) proposes a syntactic-aware text classification method that embeds weight vectors of feature words, improving recognition accuracy. Lu (2024) introduces

schema-aware hyper-relational knowledge graph embeddings for link prediction. Li (2024) presents BoostXML, a gradient boosting method for extreme multilabel text classification with tail labels, outperforming state-of-the-art baselines. Ye (2024) develops MatchXML, an efficient text-label matching framework for extreme multi-label text classification. These studies (Wang, 2025; Lu, 2024; Li, 2024; Ye, 2024) demonstrate advancements in text classification and entity alignment, with applications in various domains.

- **Education and Adaptive Systems :** (Villegas-Ch W. (2025) [[84](#)])

Villegas-Ch (2025) integrates NLP and XAI for automated semantic assessment and adaptive tutoring in Moodle.

- **Data Generation and Prediction :** (Mohanty P. (2025) [[72](#)])

Mohanty (2025) proposes TimeGAN for synthetic data generation.

Theme 4 / BERT in Cybersecurity and Beyond

Summary:

This theme encompasses a wide range of topics related to natural language processing (NLP) and machine learning (ML), with a significant focus on the application of BERT and its variants in various domains. The titles suggest applications in cybersecurity, healthcare, sustainability, and e-commerce.

Synthesis of all articles in this group:

The cluster 'BERT in Cybersecurity and Beyond' encompasses a wide range of applications of BERT and other large language models in various fields, including cybersecurity, healthcare and biomedical text analysis, text classification and fake news detection, e-commerce and consumer behavior analysis, sustainability and information extraction, and chatbots and cognitive intent analysis. In cybersecurity, studies have focused on improving threat detection and risk assessment using BERT-based models, such as the work by Ferrag (2024) and Abibulaiev (2026). In healthcare, techniques like BiLSTM-GRU and domain-specific tokenization have been utilized for sentiment analysis and medical report generation, as seen in Singh (2025) and Ashfaq (2025). Text classification and fake news detection have also been explored, with approaches like friend-guard textual adversarial examples and description-enhanced label embedding contrastive learning, as proposed by Kwon (2025) and Zhang (2024). Additionally, BERT has been applied in e-commerce for analyzing consumer behavior and predicting users' next purchases, as demonstrated by Xiong (2024) and Jeong (2024). In sustainability, BERT has been used for keyword extraction and topic modeling, with high accuracy, as shown in Gupta (2024) and Bhatt (2025). Finally, chatbots have been developed for cognitive intent analysis using Chinese BERT, as introduced by Lee (2024). These studies demonstrate the versatility and potential of BERT and other large language models in enhancing various aspects of cybersecurity, healthcare, text analysis, e-commerce, sustainability, and chatbots. Overall, the cluster highlights the significance of developing robust and accurate models for real-world applications, as emphasized by Aljawarneh (2024) and Kwon (2025).

Subthemes:

- **Cybersecurity :** (Ferrag M. (2024) [[87](#)], Abibulaiev A. (2026) [[88](#)], Lee J. (2026) [[89](#)], Yu B. (2024) [[95](#)], He D. (2024) [[100](#)], Shen H. (2026) [[5](#)])

Recent studies have focused on improving cybersecurity through the use of large language models and machine learning techniques. Ferrag (2024) proposed a privacy-preserving BERT-based lightweight model for IoT/IIoT devices, which can detect cyber threats effectively. Abibulaiev (2026) presented a context-aware ML/NLP pipeline for real-time anomaly detection and risk assessment in cloud API traffic, which can bridge the gap between ML/NLP research and practical API protection. Lee (2026) introduced a hybrid ensemble learning approach for malicious URL detection using BERT and boosting models. Yu (2024) developed a modified BERT variant, M-BERT, which can accurately classify malicious websites with a precision rate of 94.42%. He (2024) proposed a method for detecting phishing websites based on Tiny-Bert stacking, and Shen (2026) presented a split-federated BERT with adversarial training for edge intrusion detection. These studies demonstrate the potential of large language models and machine learning techniques in enhancing cybersecurity.

• **Healthcare and Biomedical Text Analysis** : (Singh D. (2025) [90], Ashfaq F. (2025) [97])

Healthcare and Biomedical Text Analysis focuses on sentiment analysis (Singh, 2025) and medical report generation (Ashfaq, 2025), utilizing techniques like BiLSTM-GRU and domain-specific tokenization to improve accuracy and coherence in healthcare decision-making.

• **Text Classification and Fake News Detection** : (Aljawarneh S. (2024) [92], Kwon H. (2025) [93], Zhang K. (2024) [99])

Text classification and fake news detection are crucial tasks in natural language processing, as noted by Aljawarneh (2024). Kwon (2025) proposes a friend-guard textual adversarial example for text classification systems, which can generate examples that are correctly classified by a friend model but misclassified by an enemy model. Zhang (2024) introduces a description-enhanced label embedding contrastive learning approach for text classification, which improves classification performance by exploiting label information. These studies highlight the importance of developing robust text classification models, as emphasized by Aljawarneh (2024) and Kwon (2025). The original title of the subtheme is: Text Classification and Fake News Detection

• **E-commerce and Consumer Behavior Analysis** : (Xiong Y. (2024) [96], Jeong B. (2024) [101])

E-commerce behavior is analyzed using text features (Xiong, 2024) and NLP-based recommendations (Jeong, 2024), enabling the exploration of consumption intention and prediction of users' next purchases, with applications in live e-commerce and product generation.

• **Sustainability and Information Extraction** : (Gupta A. (2024) [91], Bhatt P. (2025) [94])

Gupta (2024) uses BERT and YAKE for keyword extraction on sustainability reports with 98% accuracy. Bhatt (2025) assesses technology convergence using BERT topic modeling and association rule mining. Both studies leverage NLP techniques, as seen in Gupta (2024) and Bhatt (2025).

• **Chatbots and Cognitive Intent Analysis** : (Lee J. (2024) [98])

Lee (2024) proposes a chatbot for cognitive intent analysis, utilizing Chinese BERT

Theme 5 / NLP in Education and Social Media

Summary:

This theme encompasses a wide range of topics primarily focused on natural language processing (NLP) and its applications in various fields such as education, healthcare, and social media. The cluster includes titles related to text classification techniques, surveys on graph neural networks for text classification, and the use of NLP in detecting fake news and sentiment analysis.

Synthesis of all articles in this group:

The cluster 'NLP in Education and Social Media' encompasses various subthemes, including text classification techniques, NLP applications in education, sentiment analysis and fake news detection, and the integration of NLP with other technologies. According to Taha (2024), text classification techniques have been extensively surveyed, with a focus on graph-based approaches and neural architecture search methods. In education, NLP supports automatic question generation (Saleem, 2026) and creates immersive learning environments when integrated with AI and AR (Samala, 2025). Sentiment analysis and fake news detection have also gained popularity, with deep neural networks and word vector models being employed (Khan, 2024). Furthermore, NLP integrates with vision-language models for real-time safety detection (Sivanraj, 2026) and federated learning for privacy-preserving knowledge graph embeddings in IoT services (Sun, 2026). These studies demonstrate the growing interest in NLP applications across various domains, including education and social media. As noted by Yan (2025), bipartite graph-based NAS for text classification can enhance generalization capability and reduce computational complexity. Overall, the cluster highlights the potential of NLP to support various applications, from education and sentiment analysis to fake news detection and integration with other technologies.

Subthemes:

• **Text Classification Surveys and Techniques :** (Taha K. (2024) [104], Wang K. (2024) [105], Haider Rizvi S. (2025) [106], Taha K. (2025) [115], Zhang S. (2025) [110], Wang G. (2024) [112], Yan X. (2024) [113], Yan X. (2025) [114])

Recent studies have focused on text classification techniques, with Taha (2024) providing a comprehensive survey of text classification techniques and their research applications. Wang (2024) surveyed graph neural networks for text classification, discussing graph construction mechanisms and graph-based learning processes. Haider Rizvi (2025) summarized and categorized various graph convolutional network-based text classification approaches, identifying their strengths and limitations. Taha (2025) proposed a new classification system to categorize text classification algorithms into an organized structure. Zhang (2025) introduced a short text topic modeling method based on integrating Gaussian and Logistic coding networks with pre-trained word embeddings. Wang (2024) proposed a novel heterogeneous graph convolutional network method for classifying electronic health records texts. Yan (2024) presented a neural architecture search approach via multi-hashing embedding and graph tensor networks for multilingual text classification. These studies demonstrate the growing interest in text classification techniques, with a focus on graph-based approaches and neural architecture search methods. Yan (2025) proposes bipartite graph-based NAS for text classification, enhancing generalization capability and reducing computational complexity.

• **NLP Applications in Education :** (Saleem M. (2026) [108], Samala A. (2025) [116])
NLP supports automatic question generation (Saleem, 2026) and creates immersive learning environments when integrated with AI and AR (Samala, 2025).

• **NLP for Sentiment Analysis and Fake News Detection :** (Khan L. (2024) [109], Ullah K. (2025) [111], Ganesh Gopal D. (2024) [107])
Sentiment analysis and fake news detection have gained popularity, with deep neural networks (DNN) and word vector models being employed (Khan, 2024). An

attention-based stacked CNN-Bi-LSTM DNN with multilingual BERT has been proposed for Urdu sentiment analysis, achieving accuracies of 83.12% and 78.91% (Khan, 2024). Meanwhile, an ensemble-based multi-classification model has been developed for hate and offensive text classification in Urdu, achieving a macro F1 score of 0.94 (Ullah, 2025). Additionally, an AI-assisted deep NLP-based approach has been proposed for predicting fake news from social media users (Ganesh Gopal D., 2024).

- **Integration of NLP with Other Technologies :** (Sivanraj S. (2026) [102], Sun H. (2026) [103])

NLP integrates with vision-language models for real-time safety detection (Sivanraj, 2026) and federated learning for privacy-preserving knowledge graph embeddings in IoT services (Sun, 2026), enhancing entity and relationship alignment.

Theme 6 / Hierarchical Text Classification

Summary:

This theme consists of three citations focused on hierarchical text classification, a technique used in natural language processing to categorize text into multiple labels or categories. The citations explore different approaches to improve the accuracy and efficiency of hierarchical text classification.

Synthesis of all articles in this group:

Hierarchical text classification utilizes graph-based methods, including label-aware dual graph attention networks and disentangled feature graphs (Cheng, 2025; Liu, 2025). Additionally, optimization techniques such as structural entropy and singular smoothing are applied to improve performance (Liu, 2025). These approaches aim to enhance the accuracy of hierarchical text classification.

Subthemes:

- **Graph-based Methods :** (Cheng Q. (2025) [117], Liu R. (2025) [119])

Graph-based methods are used for hierarchical text classification (Cheng, 2025; Liu, 2025). Label-aware dual graph attention networks and disentangled feature graphs are applied (Cheng, 2025; Liu, 2025).

- **Optimization Techniques :** (Liu Q. (2025) [118])

Liu (2025) optimizes hierarchical text classification using structural entropy and singular smoothing.

Miscellaneous / Articles that do not have any bibliometric communality with other texts in the map:

- (Dong Y. (2026) [1], Asif M. (2025) [2], Philipo A. (2025) [3], Shen L. (2025) [6], Bozyigit F. (2024) [7], Sen A. (2024) [8])

Summary:

These articles could have some communalities with previously identified themes but communalities could not be assessed through bibliometric analyses (ARTIREV). Hence, communalities are solely assessed with the help of agentic AI (SOCRATES), who will

propose some classification that will have to be verified by the user

Synthesis of all articles in this group:

The provided citations highlight various applications of natural language processing (NLP) and deep learning in different fields. According to Asif (2025), leveraging cognitive machine reasoning and NLP can enhance network performance and efficiency. Similarly, Philipo (2025) and Shen (2025) demonstrate the effectiveness of NLP in text classification for cyberbullying detection and SMS spam detection, respectively. Dong (2026) proposes a structure-based robust fractal graph neural network for molecular property prediction. Bozyigit (2024) presents a benchmark dataset for generating domain models from natural language text using NLP. Additionally, Sen (2024) develops a software-based prototype for live event detection using NLP and deep learning to ensure people's safety. These studies showcase the potential of NLP and deep learning in addressing complex problems across various domains. The use of BERT and multi-graph convolutional networks, as seen in Shen (2025), can improve the accuracy of SMS spam detection. Furthermore, the application of NLP in automated intent-based networking, as discussed in Asif (2025), can lead to significant reductions in latency and improvements in throughput. Overall, the citations demonstrate the versatility and effectiveness of NLP and deep learning in driving innovation and solving real-world problems.

Bibliography :

- [1] : Dong Y., Xu M., Tang L., (2026), Structure-Based Robust Fractal Graph Neural Network with Molecular Fingerprint BERT for Molecular Property Prediction, IEEE Transactions on Emerging Topics in Computational Intelligence, 10, 1. [10.1109/tetci.2025.3543308](https://doi.org/10.1109/tetci.2025.3543308)
- [2] : Asif M., Ahmed Khan T., Song W., (2025), Leveraging Cognitive Machine Reasoning and NLP for Automated Intent-Based Networking and e2e Service Orchestration, IEEE Access, 13, . [10.1109/access.2025.3534282](https://doi.org/10.1109/access.2025.3534282)
- [3] : Philipo A., Sarwatt D., Ding J., Daneshmand M., Ning H., (2025), Assessing Text Classification Methods for Cyberbullying Detection on Social Media Platforms, IEEE Transactions on Information Forensics and Security, 20, . [10.1109/tifs.2025.3588728](https://doi.org/10.1109/tifs.2025.3588728)
- [4] : Lim H., Lee J., Choi D., (2026), PGB: One-Shot Pruning for BERT via Weight Grouping and Permutation, Journal of Artificial Intelligence Research, 85, . [10.1613/jair.1.20723](https://doi.org/10.1613/jair.1.20723)
- [5] : Shen H., Liu Q., Li F., Wang T., Dai Y., Bai G., (2026), Split-Federated BERT with Adversarial Training for Edge Intrusion Detection, IEEE Internet of Things Journal, , . [10.1109/jiot.2026.3679000](https://doi.org/10.1109/jiot.2026.3679000)
- [6] : Shen L., Wang Y., Li Z., Ma W., (2025), SMS spam detection using BERT and multi-graph convolutional networks, International Journal of Intelligent Networks, 6, . [10.1016/j.ijin.2025.06.002](https://doi.org/10.1016/j.ijin.2025.06.002)
- [7] : Bozyigit F., Bardakci T., Khalilipour A., Challenger M., Ramackers G., Babur Ö., Chaudron M., (2024), Generating domain models from natural language text using NLP: a benchmark dataset and experimental comparison of tools, Software and Systems Modeling, 23, 6. [10.1007/s10270-024-01176-y](https://doi.org/10.1007/s10270-024-01176-y)

- [8] : Sen A., Rajakumaran G., Mahdal M., Usharani S., Rajasekharan V., Vincent R., Sugavanan K., (2024), Live Event Detection for People's Safety Using NLP and Deep Learning, IEEE Access, 12, . [10.1109/access.2023.3349097](https://doi.org/10.1109/access.2023.3349097)
- [9] : Zhu Y., Wang Y., Qiang J., Wu X., (2024), Prompt-Learning for Short Text Classification, IEEE Transactions on Software Engineering, 36, 10. [10.1109/tkde.2023.3332787](https://doi.org/10.1109/tkde.2023.3332787)
- [10] : Aljofey A., Bello S., Lu J., Xu C., (2025), BERT-PhishFinder: A Robust Model for Accurate Phishing URL Detection With Optimized DistilBERT, IEEE Transactions on Dependable and Secure Computing, 22, 4. [10.1109/tdsc.2025.3545771](https://doi.org/10.1109/tdsc.2025.3545771)
- [11] : Song R., Li Y., Tian M., Wang H., Giunchiglia F., Xu H., (2025), Causal keyword driven reliable text classification with large language model feedback, Information Processing and Management, 62, 2. [10.1016/j.ipm.2024.103964](https://doi.org/10.1016/j.ipm.2024.103964)
- [12] : Wang Y., Zhou L., Zhang W., Zhang F., Wang Y., (2025), A soft prompt learning method for medical text classification with simulated human cognitive capabilities, Artificial Intelligence Review, 58, 4. [10.1007/s10462-025-11121-0](https://doi.org/10.1007/s10462-025-11121-0)
- [13] : Chi D., Huang T., Jia Z., Zhang S., (2025), Research on sentiment analysis of hotel review text based on BERT-TCN-BiLSTM-attention model, Array, 25, . [10.1016/j.array.2025.100378](https://doi.org/10.1016/j.array.2025.100378)
- [14] : Abdulreda Kadhim E., Feizi-Derakhshi M., Aghdasi H., (2025), Advanced Text Summarization Model Incorporating NLP Techniques and Feature-Based Scoring, IEEE Access, 13, . [10.1109/access.2025.3528830](https://doi.org/10.1109/access.2025.3528830)
- [15] : Chai X., Zhang H., Zhang J., Sun Y., Das S., (2025), Log Sequence Anomaly Detection Based on Template and Parameter Parsing via BERT, IEEE Transactions on Dependable and Secure Computing, 22, 2. [10.1109/tdsc.2024.3428538](https://doi.org/10.1109/tdsc.2024.3428538)
- [16] : Nasser M., Hamdy A., Marzouk M., (2026), Analyzing construction contract selection factors: A semantic analysis using NLP, Advanced Engineering Informatics, 69, . [10.1016/j.aei.2025.103810](https://doi.org/10.1016/j.aei.2025.103810)
- [17] : Khan W., Ishrat M., Nadeem Ahmed M., Abidin S., Husain M., Izhar M., Zamani A., Rashid Hussain M., Ali A., (2025), Enhancing Anomaly Detection in Attributed Networks Using Proximity Preservation and Advanced Embedding Techniques, IEEE Access, 13, . [10.1109/access.2025.3544260](https://doi.org/10.1109/access.2025.3544260)
- [18] : Zandbiglari K., Kumar S., Bilal M., Goodin A., Rouhizadeh M., (2025), Enhancing suicidal behavior detection in EHRs: A multi-label NLP framework with transformer models and semantic retrieval-based annotation, Journal of Biomedical Informatics, 161, . [10.1016/j.jbi.2024.104755](https://doi.org/10.1016/j.jbi.2024.104755)
- [19] : Mao X., Li Z., Li Q., Zhang S., (2025), BERT-DXLMA: Enhanced representation learning and generalization model for english text classification, Neurocomputing, 622, . [10.1016/j.neucom.2024.129325](https://doi.org/10.1016/j.neucom.2024.129325)
- [20] : Azad M., Leeon S., Khan R., Mohammed N., Momen S., (2025), SAD: Self-assessment of depression for Bangladeshi university students using machine learning and NLP, Array, 25, . [10.1016/j.array.2024.100372](https://doi.org/10.1016/j.array.2024.100372)
- [21] : Levkovich I., Omar M., (2024), Evaluating of BERT-based and Large Language Mod for Suicide Detection, Prevention, and Risk Assessment: A Systematic Review,

Journal of Medical Systems, 48, 1. [10.1007/s10916-024-02134-3](https://doi.org/10.1007/s10916-024-02134-3)

[22] : Si J., Yang J., Xiang Y., Wang H., Li L., Zhang R., Tu B., Chen X., (2024), TrajBERT: BERT-Based Trajectory Recovery With Spatial-Temporal Refinement for Implicit Sparse Trajectories, IEEE Transactions on Mobile Computing, 23, 5.

[10.1109/tmc.2023.3297115](https://doi.org/10.1109/tmc.2023.3297115)

[23] : Niu H., Omitaomu O., Langston M., Olama M., Ozmen O., Klasky H., Laurio A., Ward M., Nebeker J., (2024), EHR-BERT: A BERT-based model for effective anomaly detection in electronic health records, Journal of Biomedical Informatics, 150, .

[10.1016/j.jbi.2024.104605](https://doi.org/10.1016/j.jbi.2024.104605)

[24] : Wang Y., Zhang J., Yang Z., Wang B., Jin J., Liu Y., (2024), Improving extractive summarization with semantic enhancement through topic-injection based BERT model, Information Processing and Management, 61, 3. [10.1016/j.ipm.2024.103677](https://doi.org/10.1016/j.ipm.2024.103677)

[25] : Hussain A., Saadia A., Alserhani F., (2025), Ransomware detection and family classification using fine-tuned BERT and RoBERTa models, Egyptian Informatics Journal, 30, . [10.1016/j.eij.2025.100645](https://doi.org/10.1016/j.eij.2025.100645)

[26] : Wang B., Liang B., Bai Z., Yang M., Gui L., Xu R., (2025), GazeGCN: Gaze-aware Graph Convolutional Network for Text Classification, Neurocomputing, 611, .

[10.1016/j.neucom.2024.128680](https://doi.org/10.1016/j.neucom.2024.128680)

[27] : Ferro S., Giovanelli R., Leeson M., De Bernardin M., Traviglia A., (2025), A novel NLP-driven approach for enriching artefact descriptions, provenance, and entities in cultural heritage, Neural Computing and Applications, 37, 25.

[10.1007/s00521-025-11449-2](https://doi.org/10.1007/s00521-025-11449-2)

[28] : Hyder S., Tariq N., Moqurrah S., Ashraf M., Yoo J., Srivastava G., (2024), BERT-Based Deceptive Review Detection in Social Media: Introducing DeceptiveBERT, IEEE Transactions on Computational Social Systems, 11, 6. [10.1109/tcss.2024.3403937](https://doi.org/10.1109/tcss.2024.3403937)

[29] : Onan A., Alhumyani H., (2024), FuzzyTP-BERT: Enhancing extractive text summarization with fuzzy topic modeling and transformer networks, Journal of King Saud University - Computer and Information Sciences, 36, 6.

[10.1016/j.jksuci.2024.102080](https://doi.org/10.1016/j.jksuci.2024.102080)

[30] : Bucur A., Moldovan A., Parvatikar K., Zampieri M., Khudabukhsh A., Dinu L., (2025), On the State of NLP Approaches to Modeling Depression in Social Media: A Post-COVID-19 Outlook, IEEE Journal of Biomedical and Health Informatics, 29, 6.

[10.1109/jbhi.2025.3540507](https://doi.org/10.1109/jbhi.2025.3540507)

[31] : Rathnayake H., Sumanapala J., Rukshani R., Ranathunga S., (2024), AdapterFusion-based multi-task learning for code-mixed and code-switched text classification, Engineering Applications of Artificial Intelligence, 127, .

[10.1016/j.engappai.2023.107239](https://doi.org/10.1016/j.engappai.2023.107239)

[32] : Guleria P., Frnda J., Srinivasu P., (2025), NLP based text classification using TF-IDF enabled fine-tuned long short-term memory: An empirical analysis, Array, 27, .

[10.1016/j.array.2025.100467](https://doi.org/10.1016/j.array.2025.100467)

[33] : Su S., Shao D., Ma L., Yi S., Yang Z., (2025), ADCL: An attention feature enhancement network based on adversarial contrastive learning for short text classification, Advanced Engineering Informatics, 65, . [10.1016/j.aei.2025.103202](https://doi.org/10.1016/j.aei.2025.103202)

- [34] : Endalieu D., (2025), Fine-Tuning BERT Models for Multiclass Amharic News Document Categorization, Complexity, 2025, 1. [10.1155/cplx/1884264](https://doi.org/10.1155/cplx/1884264)
- [35] : Ferdinan T., Kocoń J., (2025), Fortifying NLP models against poisoning attacks: The power of personalized prediction architectures, Information Fusion, 114, . [10.1016/j.inffus.2024.102692](https://doi.org/10.1016/j.inffus.2024.102692)
- [36] : Chi Z., Huang H., Liu L., Bai Y., Gao X., Mao X., (2024), Can Pretrained English Language Models Benefit Non-English NLP Systems in Low-Resource Scenarios?, IEEE/ACM Transactions on Audio Speech and Language Processing, 32, . [10.1109/taslp.2023.3267618](https://doi.org/10.1109/taslp.2023.3267618)
- [37] : Ishtiaq A., Munir K., Raza A., Samee N., Jamjoom M., Ullah Z., (2024), Product Helpfulness Detection With Novel Transformer Based BERT Embedding and Class Probability Features, IEEE Access, 12, . [10.1109/access.2024.3390605](https://doi.org/10.1109/access.2024.3390605)
- [38] : Amirifar T., Lahmiri S., Zanjani M., (2024), An NLP-Deep Learning Approach for Product Rating Prediction Based on Online Reviews and Product Features, IEEE Transactions on Computational Social Systems, 11, 6. [10.1109/tcss.2023.3290558](https://doi.org/10.1109/tcss.2023.3290558)
- [39] : Wang J., Huang J., Tu X., Wang J., Huang A., Laskar M., Bhuiyan A., (2024), Utilizing BERT for Information Retrieval: Survey, Applications, Resources, and Challenges, ACM Computing Surveys, 56, 7. [10.1145/3648471](https://doi.org/10.1145/3648471)
- [40] : Yang X., Li B., Jia H., Ma Z., Sun H., Xue C., Zhang Z., (2026), A method of named entity recognition in traditional Chinese medical cases by integrating BERT and BiLSTM, Biomedical Signal Processing and Control, 112, . [10.1016/j.bspc.2025.108551](https://doi.org/10.1016/j.bspc.2025.108551)
- [41] : Wan B., Wu P., Yeo C., Li G., (2024), Emotion-cognitive reasoning integrated BERT for sentiment analysis of online public opinions on emergencies, Information Processing and Management, 61, 2. [10.1016/j.ipm.2023.103609](https://doi.org/10.1016/j.ipm.2023.103609)
- [42] : Qin L., Chen Q., Feng X., Wu Y., Zhang Y., Li Y., Li M., Che W., Yu P., (2026), Large language models meet NLP: a survey, Frontiers of Computer Science, 20, 11. [10.1007/s11704-025-50472-3](https://doi.org/10.1007/s11704-025-50472-3)
- [43] : Shehzad F., Mennella C., Esposito M., Minutolo A., (2026), Efficient multimodal learning using BERT and vision transformers for visual question answering on peripheral blood cells, Discover Artificial Intelligence, 6, 1. [10.1007/s44163-026-01011-x](https://doi.org/10.1007/s44163-026-01011-x)
- [44] : Fatemi S., Hu Y., Mousavi M., (2025), A Comparative Analysis of Instruction Fine-Tuning Large Language Models for Financial Text Classification, ACM Transactions on Management Information Systems, 16, 1. [10.1145/3706119](https://doi.org/10.1145/3706119)
- [45] : Danyal M., Khan S., Khan M., Ullah S., Ghaffar M., Khan W., (2024), Sentiment analysis of movie reviews based on NB approaches using TF-IDF and count vectorizer, Social Network Analysis and Mining, 14, 1. [10.1007/s13278-024-01250-9](https://doi.org/10.1007/s13278-024-01250-9)
- [46] : Qin S., Zhang M., (2024), Boosting generalization of fine-tuning BERT for fake news detection, Information Processing and Management, 61, 4. [10.1016/j.ipm.2024.103745](https://doi.org/10.1016/j.ipm.2024.103745)
- [47] : Galal O., Abdel-Gawad A., Farouk M., (2024), Rethinking of BERT sentence embedding for text classification, Neural Computing and Applications, 36, 32. [10.1007/s00521-024-10212-3](https://doi.org/10.1007/s00521-024-10212-3)

- [48] : Shi S., Hu K., Xie J., Guo Y., Wu H., (2024), Robust scientific text classification using prompt tuning based on data augmentation with L2 regularization, *Information Processing and Management*, 61, 1. [10.1016/j.ipm.2023.103531](https://doi.org/10.1016/j.ipm.2023.103531)
- [49] : Chen Y., Zhou T., Zeng D., Li S., Liu K., Zhao J., (2026), ASDE: Low-budget text classification via active semi-supervised learning with debiasing training mechanism, *Information Processing and Management*, 63, 2. [10.1016/j.ipm.2025.104390](https://doi.org/10.1016/j.ipm.2025.104390)
- [50] : Mueller A., Brinkmann J., Li M., Marks S., Pal K., Prakash N., Rager C., Sankaranarayanan A., Sharma A., Sun J., Todd E., Bau D., Belinkov Y., (2026), The Quest for the Right Mediator: Surveying Mechanistic Interpretability for NLP Through the Lens of Causal Mediation Analysis, *Computational Linguistics*, 52, 1. [10.1162/coli.a.572](https://doi.org/10.1162/coli.a.572)
- [51] : Alimi T., Boujelbene R., Belguith L., (2026), ARDIAL-BERT: Advancing Multidialectal Arabic Named Entity Recognition Through Continual Pretraining, *IEEE Transactions on Artificial Intelligence*, 7, 4. [10.1109/tai.2025.3562162](https://doi.org/10.1109/tai.2025.3562162)
- [52] : Bosma J., Dercksen K., Builtjes L., André R., Roest C., Fransen S., Noordman C., Navarro-Padilla M., Lefkes J., Alves N., De Grauw M., Van Eekelen L., Spronck J., Schuurmans M., De Wilde B., Hendrix W., Aswolinskiy W., Saha A., Twilt J., Geijs D., Veltman J., Yakar D., De Rooij M., Ciompi F., Hering A., Geerdink J., Huisman H., De Grauw M., Van Eekelen L., De Wilde B., Van Lohuizen Q., Stegeman M., Rutten K., Smit I., Stultiens G., Overduin C., Rutten M., Scholten E., Van Der Post R., Grünberg K., Vos S., Taken E., Nagtegaal I., Mickan A., Groeneveld M., Gerke P., Meakin J., Looijen-Salamon M., De Haas T., Hoitsma F., D'Amato M., De Rooij M., (2025), The DRAGON benchmark for clinical NLP, *npj Digital Medicine*, 8, 1. [10.1038/s41746-025-01626-x](https://doi.org/10.1038/s41746-025-01626-x)
- [53] : Zhang Y., Xu Y., Dong F., (2025), An enhanced few-shot text classification approach by integrating topic modeling and prompt-tuning, *Neurocomputing*, 617, . [10.1016/j.neucom.2024.129082](https://doi.org/10.1016/j.neucom.2024.129082)
- [54] : Alshaikh K., Almatrafi O., Abushark Y., (2024), BERT-Based Model for Aspect-Based Sentiment Analysis for Analyzing Arabic Open-Ended Survey Responses: A Case Study, *IEEE Access*, 12, . [10.1109/access.2023.3348342](https://doi.org/10.1109/access.2023.3348342)
- [55] : Boonyarat P., Liew D., Chang Y., (2024), Leveraging enhanced BERT models for detecting suicidal ideation in Thai social media content amidst COVID-19, *Information Processing and Management*, 61, 4. [10.1016/j.ipm.2024.103706](https://doi.org/10.1016/j.ipm.2024.103706)
- [56] : Cao Z., Liu X., Zhou Z., Ding L., Shang W., (2025), KD-BERT: A Lightweight Knowledge Distillation Bidirectional Encoder Representations from Transformers for IoT Network Intrusion Detection, *IEEE Transactions on Industrial Informatics*, 21, 11. [10.1109/tii.2025.3582375](https://doi.org/10.1109/tii.2025.3582375)
- [57] : Li S., Cao Y., Peng G., Li M., Sun W., Chen L., (2025), Efficient Intrusion Detection for In-Vehicle Networks Using Knowledge Distillation From BERT to CNN-BiLSTM, *IEEE Transactions on Information Forensics and Security*, 20, . [10.1109/tifs.2025.3581117](https://doi.org/10.1109/tifs.2025.3581117)
- [58] : Rostam Z., Kertész G., (2025), Advances in Pre-trained Language Models for Domain-Specific Text Classification: A Systematic Review, *ACM Transactions on Intelligent Systems and Technology*, 16, 6. [10.1145/3763002](https://doi.org/10.1145/3763002)
- [59] : Zhang L., Xia P., Ma X., Yang C., Ding X., (2024), Enhanced Chinese named entity recognition with multi-granularity BERT adapter and efficient global pointer, *Complex and Intelligent Systems*, 10, 3. [10.1007/s40747-024-01383-6](https://doi.org/10.1007/s40747-024-01383-6)

- [60] : Kumar A., Singh J., Singh A., (2025), Explainable BERT-LSTM Stacking for Sentiment Analysis of COVID-19 Vaccination, IEEE Transactions on Computational Social Systems, 12, 3. [10.1109/tcss.2023.3329664](https://doi.org/10.1109/tcss.2023.3329664)
- [61] : Anggrainingsih R., Hassan G., Datta A., (2025), Evaluating BERT-based language models for detecting misinformation, Neural Computing and Applications, 37, 16. [10.1007/s00521-025-11101-z](https://doi.org/10.1007/s00521-025-11101-z)
- [62] : Ardeshir-Larijani E., Nasiri Fatmehsari M., (2024), Hybrid classical-quantum transfer learning for text classification, Quantum Machine Intelligence, 6, 1. [10.1007/s42484-024-00147-2](https://doi.org/10.1007/s42484-024-00147-2)
- [63] : Zhang Y., Takada S., (2025), Applying LLMs to Active Learning: Toward Cost-Efficient Cross-Task Text Classification Without Manually Labeled Data, International Journal of Intelligent Systems, 2025, 1. [10.1155/int/6472544](https://doi.org/10.1155/int/6472544)
- [64] : Guridi J., Cheyre C., Yang Q., (2025), Thoughtful Adoption of NLP for Civic Participation: Understanding Differences Among Policymakers, Proceedings of the ACM on Human-Computer Interaction, 9, 2. [10.1145/3711091](https://doi.org/10.1145/3711091)
- [65] : Benavides Cesar L., Manso-Callejo M., Cira C., (2025), Methodology Based on BERT (Bidirectional Encoder Representations from Transformers) to Improve Solar Irradiance Prediction of Deep Learning Models Trained with Time Series of Spatiotemporal Meteorological Information, Forecasting, 7, 1. [10.3390/forecast7010005](https://doi.org/10.3390/forecast7010005)
- [66] : Liu H., Das A., Boltz A., Zhou D., Pinaroc D., Lease M., Lee M., (2024), Human-centered NLP Fact-checking: Co-Designing with Fact-checkers using Matchmaking for AI, Proceedings of the ACM on Human-Computer Interaction, 8, CSCW2. [10.1145/3686962](https://doi.org/10.1145/3686962)
- [67] : Wu D., Li S., Yang J., Sawan M., (2026), Neuro-BERT: Rethinking Masked Autoencoding for Self-Supervised Neurological Pretraining, IEEE Journal of Biomedical and Health Informatics, 30, 2. [10.1109/jbhi.2024.3415959](https://doi.org/10.1109/jbhi.2024.3415959)
- [68] : Siino M., Falco M., Croce D., Rosso P., (2025), Exploring LLMs Applications in Law: A Literature Review on Current Legal NLP Approaches, IEEE Access, 13, . [10.1109/access.2025.3533217](https://doi.org/10.1109/access.2025.3533217)
- [69] : De Santis E., Martino A., Ronci F., Rizzi A., (2025), From Bag-of-Words to Transformers: A Comparative Study for Text Classification in Healthcare Discussions in Social Media, IEEE Transactions on Emerging Topics in Computational Intelligence, 9, 1. [10.1109/tetci.2024.3423444](https://doi.org/10.1109/tetci.2024.3423444)
- [70] : Rezayi S., Liu Z., Wu Z., Dhakal C., Ge B., Dai H., Mai G., Liu N., Zhen C., Liu T., Li S., (2025), Exploring New Frontiers in Agricultural NLP: Investigating the Potential of Large Language Models for Food Applications, IEEE Transactions on Big Data, 11, 3. [10.1109/tbdata.2024.3442542](https://doi.org/10.1109/tbdata.2024.3442542)
- [71] : Wang Y., Wang Q., Yu H., Xie J., Hu F., Wang X., Lei D., (2026), GCFA: Generative class feature fusion with agent attention for medical text classification, Information Fusion, 126, . [10.1016/j.inffus.2025.103639](https://doi.org/10.1016/j.inffus.2025.103639)
- [72] : Mohanty P., Jena P., Padhy N., (2025), TimeGAN-Based Diversified Synthetic Data Generation Following BERT-Based Model for EV Battery SOC Prediction: A State-of-The-Art Approach, IEEE Transactions on Industry Applications, 61, 3.

[10.1109/tia.2025.3534165](https://doi.org/10.1109/tia.2025.3534165)

[73] : Zhao Z., Meng F., Lyu Z., Li H., Li X., Zhu G., (2026), CSI-BERT2: A BERT-Inspired Framework for Efficient CSI Prediction and Classification in Wireless Communication and Sensing, IEEE Transactions on Mobile Computing, 25, 5.

[10.1109/tmc.2025.3640420](https://doi.org/10.1109/tmc.2025.3640420)

[74] : Bai L., Song X., Zhu L., (2025), Joint Multi-Feature Information Entity Alignment for Cross-Lingual Temporal Knowledge Graph With BERT, IEEE Transactions on Big Data, 11, 2. [10.1109/tbdata.2024.3378113](https://doi.org/10.1109/tbdata.2024.3378113)

[75] : Zhao X., An Y., Xu N., Geng X., (2024), Variational Continuous Label Distribution Learning for Multi-Label Text Classification, IEEE Transactions on Software Engineering, 36, 6. [10.1109/tkde.2023.3323401](https://doi.org/10.1109/tkde.2023.3323401)

[76] : Wang H., Ouyang Y., Wu Y., Jiang C., Jin L., Cao Y., Li Q., (2025), KMTLabeler: An Interactive Knowledge-Assisted Labeling Tool for Medical Text Classification, IEEE Transactions on Visualization and Computer Graphics, 31, 9.

[10.1109/tvcg.2024.3406387](https://doi.org/10.1109/tvcg.2024.3406387)

[77] : Ramakrishnan S., Dhinesh Babu L., (2025), Improving Multi-Label Emotion Classification on Imbalanced Social Media Data With BERT and Clipped Asymmetric Loss, IEEE Access, 13, . [10.1109/access.2025.3557091](https://doi.org/10.1109/access.2025.3557091)

[78] : Zhang H., Kassim A., Samsudin N., Teng L., Tang C., (2025), A Hybrid Deep Learning Framework for Hotel Rating Systems: Integrating Word2Vec, TF-IDF, and Bi-LSTM With Attention Mechanism, IEEE Transactions on Computational Social Systems, 12, 5. [10.1109/tcss.2024.3461796](https://doi.org/10.1109/tcss.2024.3461796)

[79] : Wang M., Kim J., Yan Y., (2025), Syntactic-Aware Text Classification Method Embedding the Weight Vectors of Feature Words, IEEE Access, 13, .

[10.1109/access.2025.3545877](https://doi.org/10.1109/access.2025.3545877)

[80] : Hu Y., Kim H., Ye K., Lu N., (2024), BERT-PIN: A BERT-Based Framework for Recovering Missing Data Segments in Time-Series Load Profiles, IEEE Transactions on Industrial Informatics, 20, 10. [10.1109/tii.2024.3417272](https://doi.org/10.1109/tii.2024.3417272)

[81] : Lu Y., Yang D., Wang P., Rosso P., Cudre-Mauroux P., (2024), Schema-Aware Hyper-Relational Knowledge Graph Embeddings for Link Prediction, IEEE Transactions on Software Engineering, 36, 6. [10.1109/tkde.2023.3323499](https://doi.org/10.1109/tkde.2023.3323499)

[82] : Peled-Cohen L., Reichart R., (2025), A Systematic Review of NLP for Dementia: Tasks, Datasets, and Opportunities, Transactions of the Association for Computational Linguistics, 13, . [10.1162/tacl.a.35](https://doi.org/10.1162/tacl.a.35)

[83] : Ye H., Xie X., Xie F., Zuo J., Bu C., (2025), LoRA-Adv: Boosting Text Classification in Large Language Models Through Adversarial Low-Rank Adaptations, IEEE Access, 13, . [10.1109/access.2025.3579539](https://doi.org/10.1109/access.2025.3579539)

[84] : Villegas-Ch W., Gutierrez R., García-Ortiz J., Guevara V., (2025), Explainable educational assistant integrated in Moodle: automated semantic assessment and adaptive tutoring based on NLP and XAI, Discover Artificial Intelligence, 5, 1. [10.1007/s44163-025-00438-y](https://doi.org/10.1007/s44163-025-00438-y)

[85] : Li F., Zuo Y., Lin H., Wu J., (2024), BoostXML: Gradient Boosting for Extreme Multilabel Text Classification With Tail Labels, IEEE Transactions on Neural Networks and Learning Systems, 35, 11. [10.1109/tnnls.2023.3285294](https://doi.org/10.1109/tnnls.2023.3285294)

- [86] : Ye H., Sunderraman R., Ji S., (2024), MatchXML: An Efficient Text-Label Matching Framework for Extreme Multi-Label Text Classification, IEEE Transactions on Software Engineering, 36, 9. [10.1109/tkde.2024.3374750](https://doi.org/10.1109/tkde.2024.3374750)
- [87] : Ferrag M., Ndhlovu M., Tihanyi N., Cordeiro L., Debbah M., Lestable T., Thandi N., (2024), Revolutionizing Cyber Threat Detection With Large Language Models: A Privacy-Preserving BERT-Based Lightweight Model for IoT/IIoT Devices, IEEE Access, 12, . [10.1109/access.2024.3363469](https://doi.org/10.1109/access.2024.3363469)
- [88] : Abibulaiev A., Pukach P., Vovk M., (2026), Context-Aware ML/NLP Pipeline for Real-Time Anomaly Detection and Risk Assessment in Cloud API Traffic, Machine Learning and Knowledge Extraction, 8, 1. [10.3390/make8010025](https://doi.org/10.3390/make8010025)
- [89] : Lee J., Kwon H., (2026), Hybrid Ensemble Learning for Malicious URL Detection With BERT and Boosting Models, IEEE Access, 14, . [10.1109/access.2025.3605302](https://doi.org/10.1109/access.2025.3605302)
- [90] : Singh D., Shivprakash Barve S., Krishna Dwivedi A., (2025), OptiASAR: Optimized Aspect-Based Sentiment Analysis of Reviews With BiLSTM-GRU and NER-BERT in Healthcare Decision-Making, IEEE Access, 13, . [10.1109/access.2025.3549303](https://doi.org/10.1109/access.2025.3549303)
- [91] : Gupta A., Chadha A., Tewari V., (2024), A Natural Language Processing Model on BERT and YAKE Technique for Keyword Extraction on Sustainability Reports, IEEE Access, 12, . [10.1109/access.2024.3352742](https://doi.org/10.1109/access.2024.3352742)
- [92] : Aljawarneh S., Swedat S., (2024), Fake News Detection Using Enhanced BERT, IEEE Transactions on Computational Social Systems, 11, 4. [10.1109/tcss.2022.3223786](https://doi.org/10.1109/tcss.2022.3223786)
- [93] : Kwon H., (2025), Friend-Guard Textfooler Attack on Text Classification System, IEEE Access, 13, . [10.1109/access.2021.3080680](https://doi.org/10.1109/access.2021.3080680)
- [94] : Bhatt P., Hsu Y., Lai K., Drave V., (2025), Technology Convergence Assessment by an Integrated Approach of BERT Topic Modeling and Association Rule Mining, IEEE Transactions on Engineering Management, 72, . [10.1109/tem.2025.3556006](https://doi.org/10.1109/tem.2025.3556006)
- [95] : Yu B., Tang F., Ergu D., Zeng R., Ma B., Liu F., (2024), Efficient Classification of Malicious URLs: M-BERT—A Modified BERT Variant for Enhanced Semantic Understanding, IEEE Access, 12, . [10.1109/access.2024.3357095](https://doi.org/10.1109/access.2024.3357095)
- [96] : Xiong Y., Wei N., Qiao K., Li Z., Li Z., (2024), Exploring Consumption Intent in Live E-Commerce Barrage: A Text Feature-Based Approach Using BERT-BiLSTM Model, IEEE Access, 12, . [10.1109/access.2024.3399095](https://doi.org/10.1109/access.2024.3399095)
- [97] : Ashfaq F., Jhanjhi N., Khan N., Javed D., Masud M., Shorfuzzaman M., (2025), Enhancing ECG Report Generation With Domain-Specific Tokenization for Improved Medical NLP Accuracy, IEEE Access, 13, . [10.1109/access.2025.3567566](https://doi.org/10.1109/access.2025.3567566)
- [98] : Lee J., Wu E., Ou Y., Lee Y., Lee C., Chung C., (2024), Anti-Drugs Chatbot: Chinese BERT-Based Cognitive Intent Analysis, IEEE Transactions on Computational Social Systems, 11, 1. [10.1109/tcss.2023.3238477](https://doi.org/10.1109/tcss.2023.3238477)
- [99] : Zhang K., Wu L., Lv G., Chen E., Ruan S., Liu J., Zhang Z., Zhou J., Wang M., (2024), Description-Enhanced Label Embedding Contrastive Learning for Text Classification, IEEE Transactions on Neural Networks and Learning Systems, 35, 10. [10.1109/tnnls.2023.3282020](https://doi.org/10.1109/tnnls.2023.3282020)

- [100] : He D., Lv X., Zhu S., Chan S., Choo K., (2024), A Method for Detecting Phishing Websites Based on Tiny-Bert Stacking, IEEE Internet of Things Journal, 11, 2. [10.1109/jiot.2023.3292171](https://doi.org/10.1109/jiot.2023.3292171)
- [101] : Jeong B., Lee K., (2024), NLP-Based Recommendation Approach for Diverse Service Generation, IEEE Access, 12, . [10.1109/access.2024.3355546](https://doi.org/10.1109/access.2024.3355546)
- [102] : Sivanraj S., Uduwage D., Tripathi M., (2026), Real-time safety detection on construction sites using a vision-language and NLP-based model, Advanced Engineering Informatics, 69, . [10.1016/j.aei.2025.103889](https://doi.org/10.1016/j.aei.2025.103889)
- [103] : Sun H., Bi X., Tu Z., Zhao B., Zhang K., Chu D., Xu X., (2026), Enhancing Privacy-Preserving Knowledge Graph Embeddings with Federated Learning for IoT Services, ACM Transactions on Internet Technology, 26, 1. [10.1145/3747351](https://doi.org/10.1145/3747351)
- [104] : Taha K., Yoo P., Yeun C., Homouz D., Taha A., (2024), A comprehensive survey of text classification techniques and their research applications: Observational and experimental insights, Computer Science Review, 54, . [10.1016/j.cosrev.2024.100664](https://doi.org/10.1016/j.cosrev.2024.100664)
- [105] : Wang K., Ding Y., Han S., (2024), Graph neural networks for text classification: a survey, Artificial Intelligence Review, 57, 8. [10.1007/s10462-024-10808-0](https://doi.org/10.1007/s10462-024-10808-0)
- [106] : Haider Rizvi S., Imran R., Mahmood A., (2025), Text Classification Using Graph Convolutional Networks: A Comprehensive Survey, ACM Computing Surveys, 57, 8. [10.1145/3714456](https://doi.org/10.1145/3714456)
- [107] : Ganesh Gopal D., Nagarajan S., Amanullah S., Mary S., Bashir A., (2024), AI-Assisted Deep NLP-Based Approach for Prediction of Fake News From Social Media Users, IEEE Transactions on Computational Social Systems, 11, 4. [10.1109/tcss.2023.3259480](https://doi.org/10.1109/tcss.2023.3259480)
- [108] : Saleem M., Rehman Z., Ali R., Akmal U., Ijaz A., Abbas R., (2026), An information processing framework for education: Supporting automatic question generation with NLP to minimize human intervention, Information Processing and Management, 63, 4. [10.1016/j.ipm.2026.104613](https://doi.org/10.1016/j.ipm.2026.104613)
- [109] : Khan L., Qazi A., Chang H., Alhajlah M., Mahmood A., (2024), Empowering Urdu sentiment analysis: an attention-based stacked CNN-Bi-LSTM DNN with multilingual BERT, Complex and Intelligent Systems, 11, 1. [10.1007/s40747-024-01631-9](https://doi.org/10.1007/s40747-024-01631-9)
- [110] : Zhang S., Xu J., Hui N., Zhai P., (2025), A short text topic modeling method based on integrating Gaussian and Logistic coding networks with pre-trained word embeddings, Neurocomputing, 616, . [10.1016/j.neucom.2024.128941](https://doi.org/10.1016/j.neucom.2024.128941)
- [111] : Ullah K., Aslam M., Khan M., Alamri F., Khan A., (2025), UEF-HOCUrdu: Unified Embeddings Ensemble Framework for Hate and Offensive Text Classification in Urdu, IEEE Access, 13, . [10.1109/access.2025.3532611](https://doi.org/10.1109/access.2025.3532611)
- [112] : Wang G., Lou X., Guo F., Kwok D., Cao C., (2024), EHR-HGCN: An Enhanced Hybrid Approach for Text Classification Using Heterogeneous Graph Convolutional Networks in Electronic Health Records, IEEE Journal of Biomedical and Health Informatics, 28, 3. [10.1109/jbhi.2023.3346210](https://doi.org/10.1109/jbhi.2023.3346210)
- [113] : Yan X., Huang H., Jin Y., Chen L., Liang Z., Hao Z., (2024), Neural Architecture Search via Multi-Hashing Embedding and Graph Tensor Networks for Multilingual Text Classification, IEEE Transactions on Emerging Topics in Computational Intelligence, 8, 1. [10.1109/tetci.2023.3301774](https://doi.org/10.1109/tetci.2023.3301774)

- [114] : Yan X., Huang H., Jin Y., Wang Z., Hao Z., (2025), Neural Architecture Search Based on Bipartite Graphs for Text Classification, IEEE Transactions on Neural Networks and Learning Systems, 36, 6. [10.1109/tnnls.2024.3514708](https://doi.org/10.1109/tnnls.2024.3514708)
- [115] : Taha K., Yoo P., Yeun C., Taha A., (2025), Text Classification Techniques: A Holistic Review, Observational Analysis, and Experimental Investigation, Big Data Mining and Analytics, 8, 3. [10.26599/bdma.2024.9020092](https://doi.org/10.26599/bdma.2024.9020092)
- [116] : Samala A., Rawas S., (2025), Triadic fusion: investigating the educational impact of AI, AR, and NLP integration in creating immersive and conversational learning environments, Interactive Learning Environments, , . [10.1080/10494820.2025.2554987](https://doi.org/10.1080/10494820.2025.2554987)
- [117] : Cheng Q., Shi W., (2025), Hierarchical multi-label text classification of tourism resources using a label-aware dual graph attention network, Information Processing and Management, 62, 1. [10.1016/j.ipm.2024.103952](https://doi.org/10.1016/j.ipm.2024.103952)
- [118] : Liu Q., Peng H., Huang X., Hao Z., Sun Q., Yu Z., Yu P., (2025), Hierarchical Text Classification Optimization via Structural Entropy and Singular Smoothing, IEEE Transactions on Software Engineering, 37, 9. [10.1109/tkde.2025.3579810](https://doi.org/10.1109/tkde.2025.3579810)
- [119] : Liu R., Zhang X., Yuan L., Zhou X., (2025), Disentangled feature graph for Hierarchical Text Classification, Information Processing and Management, 62, 3. [10.1016/j.ipm.2025.104065](https://doi.org/10.1016/j.ipm.2025.104065)